Chapter 1 Introduction

1.1 Motivation:

- Limited Access to Mental Health Support

Many people who need psychological support face difficulties in accessing mental health professionals due to:

- Limited availability of psychiatrists
- Geographical distance
- High cost
- Hesitation to seek psychological help

Calma provides an integrated platform enabling rapid access via a mobile application (Flutter) or a web platform (React), with AI-powered chatbot assistance (text and voice) and direct communication with approved psychiatrists.

- Lack of Continuous Psychological Monitoring

Patients' psychological conditions and moods can change over time, but traditional support methods may not provide continuous monitoring.

Calma provides a mood-tracking system, daily positive challenges, and analytical reports (weekly / monthly) to help patients track emotional changes over time.

- Difficulty Detecting Psychological Crisis Situations

Critical statements related to self-harm or other dangerous situations, whether typed or spoken, may require immediate attention. Delayed detection can lead to:

- Lack of immediate intervention
- Increased risk to the patient

Calma implements an AI-driven NLP mechanism that detects high-risk statements in real time, in both text messages and transcribed voice messages, and instantly triggers a non-dismissible red “SOS Screen” that displays emergency support numbers and prevents the user from navigating back.

- Difficulty Expressing Feelings Through Writing

Some patients find it hard to describe their emotions in writing, especially when they are anxious, exhausted, or in distress. Typing long messages can become a barrier to seeking help.

Calma supports voice interaction with the chatbot: the patient records a voice message, the system converts the speech to text, analyzes it (including self-harm detection), and returns an empathetic reply, making the conversation more natural and accessible.

- Need for Reliable Self-Care Resources

Patients require a safe environment with accessible self-help tools.

Calma provides a static library of calming sounds and utilizes an automated web-scraping data pipeline to fetch and categorize the latest mental health articles and books from trusted global sources.

- Lack of Awareness and Fear of Stigma

Many people lack reliable psychological awareness, and the fear of social judgment prevents them from sharing their struggles or learning from the experiences of others.

Calma includes a community awareness module in which verified doctors publish educational posts and practical tips, and patients can read them and benefit from them. Patients can also share their personal experiences to help others, with posts published anonymously (without displaying the patient's name) to guarantee complete privacy.

- Ensuring Psychiatrist Reliability

To protect patients, the credentials of mental health professionals must be rigorously verified.

Calma enforces a strict doctor on-boarding workflow requiring the upload of official documents (ID and graduation certificate), keeping the account in a “pending” state until it is reviewed and approved by the system Admin. Only approved doctors can appear in the doctors list and publish posts.

1.2 Project Objectives:

The main objective of this project is to develop a comprehensive, secure mental health support platform supporting three distinct user roles (Patient, Doctor, and Admin) with strict authorization. The platform aims to provide a smart AI chatbot, available through both text and voice, capable of generating empathetic responses and accurately detecting self-harm intent to trigger emergency SOS protocols. It also seeks to implement a robust mood-tracking system supported by interactive charts and positive challenges, and to build an automated data pipeline to gather psychological articles.

In addition, the platform aims to build a community awareness space in which approved doctors publish educational posts and advice, and patients read them and share their own experiences anonymously to preserve their privacy. Finally, it seeks to develop a comprehensive Admin Dashboard to manage system statistics and rigorously review doctor verification requests in order to maintain service quality.

1.3 Project Developing Methodology:

The Agile development methodology is well-suited for developing our mental health support platform. Here are some reasons why:

1. Flexibility: The Agile approach is highly flexible, allowing the development team to adapt to changing requirements and priorities quickly. This is especially useful for our project, which includes multiple components and technologies that need continuous improvement.

2. Iterative Development: The iterative development process in Agile allows each feature to be developed, tested, and refined in short cycles. This helps the team identify problems early and ensure that the platform meets the required needs.

3. Collaboration: The Agile approach emphasizes communication and collaboration between team members, which is important for coordinating the development of the mobile application, web dashboard, backend, and AI-based features (chatbot, speech-to-text, and self-harm detection).

4. Continuous Improvement: The Agile methodology encourages continuous testing, feedback, and improvement throughout the development process. This helps the team enhance the platform's quality, usability, and reliability.



Figure 1.1: Agile Development Cycle

1.4 Tools and Technologies:

1.4.1 IDEs used in the platform

IDE Name	Description	Logo
Visual Studio Code	Visual Studio Code is a free code editor that supports multiple programming languages and extensions.	
Android Studio	Android Studio is the official IDE for developing Android applications.	
Laravel	Laravel is a PHP web application framework used for developing and managing web projects and RESTful APIs.	
XAMPP	XAMPP is a local server package used to run and test web applications on a computer.	
Postman	Postman is a tool used to test APIs and send HTTP requests.	
IntelliJ	IntelliJ IDEA provides tools to run unit tests, detect errors, and verify code functionality.	

1.4.2 Definitions, Acronyms, and Abbreviations:

- **API :** Application Programming Interface
- **NLP :** Natural Language Processing
- **BERT :** Bidirectional Encoder Representations from Transformers
- **SOS :** Emergency distress alert (the red screen that displays emergency support numbers)
- **STT :** Speech-to-Text
- **DFD :** Data Flow Diagram
- **ERD :** Entity-Relationship Diagram
- **JWT :** JSON Web Token

- **RSS :** Really Simple Syndication
- **UI/UX :** User Interface / User Experience

1.4.3 Tools and Technologies

- **Mobile Development (Patient App):**

- Flutter for developing the patient mobile application.
- Dart programming language.
- fl_chart library for interactive mood report charts.
- Audio recording package (e.g., record) for capturing voice messages in the chatbot.

- **Web Development (Doctor & Admin Dashboards):**

- React.js for developing the psychiatrist and administrator web dashboards.
- Tailwind CSS for styling and responsive user interfaces.
- Recharts or Chart.js for data visualization.
- Redux or Context API for state management, including SOS screen triggers.
- Browser MediaRecorder API for voice recording on the web platform.

- **Backend & Database:**

- PHP and Laravel for developing the RESTful backend APIs.
- MySQL for database management, including the posts, comments, and anonymous-post tables.
- Laravel Sanctum for token issuance and multi-role authentication.
- Laravel Storage (or AWS) for securing sensitive uploaded documents and voice recordings.
- Firebase Cloud Messaging (FCM) or Pusher for real-time notifications.

- **AI & Data Engineering:**

- Python (FastAPI/Flask) microservices for hosting AI models.
- BERT NLP model for intent classification, self-harm detection, and sentiment analysis.
- Speech-to-Text (STT) model/service (e.g., OpenAI Whisper) to transcribe Arabic and English voice messages into text before analysis.
- Python Airflow or Cron Jobs for automated article scraping.

- **Content & Community Module:**

- Posts management API allowing approved doctors to publish awareness posts and tips.
- Anonymous patient posts: the author's identity is stored internally for moderation but is never displayed or returned to other users.
- Admin moderation tools to review, hide, or delete posts.

- **Testing:**

- PHPUnit for backend feature testing.

- Jest for frontend React testing.